

Faculty OS — Project Documentation

Comprehensive Technical & Functional Blueprint for Next-Gen Higher Education Management

1. Executive Summary

Faculty OS is an AI-powered academic management platform engineered to streamline, automate, and digitize the daily professional workflows of engineering college educators. Built specifically for higher education institutions (tailored for Anna University affiliated colleges like Sri Eshwar College of Engineering), Faculty OS unifies student mentoring, assessment generation, research tracking, academic planning, and institutional resource management into a single intuitive digital workspace.

2. System Architecture & Tech Stack

Faculty OS follows a modern, decoupled web architecture separating the user interface from backend services and AI agent orchestrators.

Layer	Technologies Used
Frontend Framework	React 19, TypeScript, Vite
Styling & UI	Tailwind CSS, Framer Motion, Lucide React Icons
Backend API	Python 3.11, FastAPI, Uvicorn
Database / ORM	SQLite, SQLAlchemy, Pydantic
AI & Automation	Web Speech API, Custom Multi-Agent Orchestrator, Gemini API
Document Exporting	jsPDF, jsPDF-AutoTable, SheetJS (XLSX)

3. Functional Module Breakdown (Phases 1 – 10)

Phase 1: Dashboard & Command Center

Real-time command dashboard offering visibility into academic alerts, pending tasks, student attendance metrics, and quick action shortcuts.

Phase 2: Student Mentoring Roster

Mentee management database displaying academic profiles, historical performance, risk categories, and contact information with Excel/CSV support.

Phase 3: Mentor Check-In & Voice Logging

Interaction log for 1-on-1 student mentoring sessions with integrated hands-free voice-to-text recording and automated meeting summary generation.

Phase 4: Mentor Wellbeing Tracker

Emotional and academic health tracker for assigned mentees featuring mood tags, health score indicators, and intervention triggers.

Phase 5: Exam & Assessment Generator

Automated NBA-standard question paper generator with Bloom's Taxonomy alignment, interactive header inline editing, dynamic logo integration, and print PDF export.

Phase 6: Research & Grants Manager

Scholarly activity manager for faculty research, IEEE/Scopus publication logging, grant application tracking, and funding deadline reminders.

Phase 7: Academic Workflow & Course Management

Planning tool for syllabus completion, lesson plan tracking, CO-PO attainment calculations, and internal mark logs.

Phase 8: Placement & Internships Coordinator

Placement drive scheduler, eligibility filtering, offer letter verification, and student internship tracking.

Phase 9: Alumni Relations & Events

Alumni directory management, guest lecture scheduling, event registration, and committee coordination.

Phase 10: Inventory & Resource Management

Tracking system for lab hardware (GPU servers), software keys, library books, and equipment requisitions.

4. AI Assistant Integration (ARIA)

Faculty OS features a **Faculty AI Assistant (ARIA — Academic Resource & Insight Assistant)** powered by a custom multi-agent engine:

- **Natural Language Querying:** Answers curriculum regulations, student statistics, and institutional guidelines.
- **Automated Document Generation:** Drafts recommendation letters, meeting notes, and official communications.
- **NBA Rubric Generator:** Builds evaluation rubrics based on question paper structures and Bloom's Taxonomy.

5. Security & Data Management

- **Role-Based Access Control (RBAC):** Restricts student record access based on faculty credentials.
- **Dual-Layer Persistence:** Combines SQLite backend storage with browser localStorage for seamless offline resiliency.
- **Data Export Compliance:** Supports XLSX and PDF formats for institutional reporting.

6. Installation & Quick Start

Frontend Setup:

```
cd frontend
npm install
npm run dev
```

Backend Setup:

```
cd backend
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
pip install -r requirements.txt
uvicorn main:app --reload
```